RN-DPE-RDHEI: Reversible Data Hiding in Encrypted Images Using Residual CNN Predictor, BRBCC Compression, and Chaos-Based Difference-Preserving Encryption

Shiva Kumar Nangunuri
Assistant Professor
Department of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
shivakumar760@gmail.com

Bassa Srilakshmi
Department of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
bassasrilakshmi58@gmail.com

***Abstract*—Reversible Data Hiding in Encrypted Images (RDHEI) allows a data hider to embed a secret payload inside a content-encrypted image without seeing the original plaintext, while a privileged receiver recovers both the payload and the original image with zero distortion. Every published Reserve Room Before Encryption (RRBE) scheme relies on hand-crafted predictors — MED, GAP, or ISGAP — whose algebraic formulas impose a hard accuracy ceiling that directly caps the achievable embedding capacity. We present RN-DPE-RDHEI, a five-component framework that breaks this ceiling. A four-layer residual convolutional neural network (CNN), trained to correct Median Edge Detector (MED) prediction residuals, reduces mean absolute error from 1.2851 to 0.0516 pixels on the BOSSBase 1.01 benchmark — a 96% improvement that directly doubles the free bit-planes available under the BRBCC block-level compression scheme. A block-wise Difference-Preserving Cipher (DPC), seeded by Lorenz ordinary differential equations and the Logistic Sine System, keyed via ECDH X25519 with HKDF-SHA256, ensures intra-block pixel differences survive encryption so BRBCC labels remain valid in the ciphertext domain. Dual HMAC-SHA256 and ECDSA Ed25519 authentication — the first such combination in any RRBE scheme — completes the framework. Experiments on 1,000 BOSSBase test images confirm PSNR $= \infty$, SSIM $= 1.0$, BPP $= 5.8327$, NPCR $= 99.60\%$, UACI $= 46.20\%$, and MAE $= 0.0516$ pixels, with 1,000/1,000 perfect image recovery, secret extraction, and authentication, and zero failures, surpassing all fifteen surveyed RDHEI papers published between 2008 and 2026.**

***Index Terms*—Reversible data hiding, encrypted images, residual CNN predictor, BRBCC compression, ECDH key exchange, chaos-based encryption, difference-preserving cipher, HMAC, ECDSA.**

I. Introduction

The rapid growth of cloud-hosted medical records, satellite imagery pipelines, and multimedia distribution networks has produced a specific security requirement that standard encryption alone cannot satisfy. An intermediary storing or forwarding encrypted content may need to annotate that content — attaching a routing identifier, a quality score, or an authentication token — without ever obtaining access to the underlying plaintext. Conventional encryption prohibits such interactions entirely.

Reversible Data Hiding in Encrypted Images (RDHEI) [1], [2] addresses this gap through a three-party protocol. The *content owner* (Alice) encrypts an image and passes the ciphertext to a *data hider*, who embeds a secret payload using only a data-hiding key. The *receiver* (Charlie), who holds both the data-hiding key and the decryption key, extracts the payload and reconstructs the original image with zero distortion — PSNR $= \infty$ and SSIM $= 1.0$. Clinical imaging archives, intelligence reconnaissance workflows, and tamper-evident digital forensics chains are among the natural application domains.

A. Architectural Overview

Two architectural families divide the literature. *Vacate Room After Encryption* (VRAE) [1] vacates embedding space inside the ciphertext by modifying a subset of encrypted bits. The practical capacity is limited to under 1 bpp because the amount of room that can be vacated without harming reversibility is small. *Reserve Room Before Encryption* (RRBE) [2] preprocesses the original image to identify and clear free bit-planes before encryption is applied, then writes secret data into those planes inside the ciphertext. RRBE systems consistently achieve 1.5 to 6 bpp and dominate recent high-capacity RDHEI research.

B. The Prediction Accuracy Bottleneck

In RRBE, embedding capacity is set entirely by prediction accuracy. The BRBCC scheme [6] formalises this relationship: it classifies each 4×4 image block according to the maximum absolute prediction error within the block and releases the corresponding bit-planes for embedding. When the predictor achieves MAE ≈ 4 pixels, three to four planes are freed per block and maximum BPP is approximately 3.9. When MAE drops below 0.1 pixel, six planes are freed and maximum BPP rises to approximately 5.96.

Despite this well-understood relationship, every published RRBE predictor to date uses a hand-crafted algebraic formula. MED [3] applies the JPEG-LS edge-detection rule. GAP exploits gradient direction. ISGAP [5] introduces seven specialised formulas for different local gradient configurations. None of these methods can learn from training data or adapt beyond the expressive power of their fixed equations. A compact convolutional network trained as a *residual corrector* on top of MED — predicting the error that MED makes rather than the pixel value itself — can reduce MAE by an order of magnitude and thereby dramatically expand the free bit-planes that BRBCC can identify.

C. Security Gaps

Prior RDHEI schemes derive encryption keys from image-specific hashes, creating a chosen-plaintext vulnerability: an attacker who obtains the original image can reconstruct the key [12]. ECDH X25519 key exchange, whose security rests on elliptic-curve key material rather than image content, eliminates this vulnerability. Additionally, while ISGAP [5] introduced HMAC-SHA256 for ciphertext integrity, no published RRBE system pairs HMAC with a digital signature for sender non-repudiation.

D. Contributions

1) **First residual CNN predictor in RDHEI.** A four-layer network reduces MAE from 1.2851 to 0.0516 px — 96% below MED and 98% below ISGAP [5], the best prior result.
2) **Highest BPP on BOSSBase 1.01.** BPP = 5.8327 on 1,000 test images, a 48% improvement over BRBCC+MED [6] (3.9381).
3) **Image-independent key exchange.** ECDH X25519 with HKDF-SHA256 removes the chosen-plaintext vulnerability present in all prior RDHEI key management schemes.
4) **First dual HMAC + ECDSA in RRBE.** HMAC-SHA256 for data integrity and ECDSA Ed25519 for sender non-repudiation, combined for the first time in any RRBE-based RDHEI paper.
5) **Verified real-world three-party pipeline.** Only cryptographic keys and image files are exchanged; no auxiliary data files are needed at any step.
6) **Comprehensive zero-failure evaluation.** Twelve standard metrics measured on 1,000 images with zero failures — the most thorough published evaluation of any RRBE system.

II. Related Work

A. Foundational Frameworks

Zhang [1] introduced RDHEI by extending the LSB-substitution steganography paradigm to the encrypted domain, achieving ≈ 0.1 bpp via VRAE and establishing the three-party protocol structure that all subsequent work inherits. Ma et al. [2] made the pivotal architectural shift to RRBE, demonstrating that reserving embedding space before encryption raises capacity to ≈ 1.5 bpp while guaranteeing perfect image reconstruction. HC-RDHEI [9] extended RRBE with multi-bit error prediction embedding, reaching 1.836 bpp, and provided the first explicit quantification of the predictor-capacity coupling.

B. Predictor-Based Capacity Methods

Combining MED with GAP and adaptive Huffman coding [3] reached 3.877 bpp by adapting the predictor choice to local gradient information. ACNNP [4] introduced a convolutional network as a standalone predictor with adaptive mean coding, reporting 3.461 bpp with MAE ≈ 2.8 pixels. Although ACNNP represents the first CNN-based approach in RDHEI, it targets the full pixel value rather than the MED residual, leaving a substantial accuracy gap. ISGAP [5] proposed seven gradient-based rules tailored to different local neighbourhood configurations, achieving 3.390 bpp with MAE ≈ 3.0 pixels and adding HMAC-SHA256 authentication.

C. Compressor-Based Methods

BRBCC [6] established the strongest prior result of 3.9381 bpp by pairing a rigorous block-level classification scheme with bit-plane compression. LBE+IBBE [10] reached 3.195 bpp using adaptive linear binary embedding. MSB+ERLE [11] applied MSB block labelling with enhanced run-length encoding for 3.790 bpp.

D. Encryption and Security Extensions

He et al. [12] paired a chaotic cipher with ECC-based key exchange, confirming that image-independent key establishment is achievable, but reported no improvement in embedding capacity because the underlying predictor remained hand-crafted. No prior RRBE paper simultaneously provides image-independent key exchange, ciphertext integrity via HMAC, and sender authentication via ECDSA.

Table I consolidates all methods. RN-DPE-RDHEI is the first single framework to close all identified gaps.

TABLE I
COMPARISON OF RDHEI METHODS ON BOSSBASE 1.01. "–" = NOT REPORTED.

Method	BPP	MAE	Auth.	Year
Zhang [1]	0.100	–	None	2011
Ma [2]	1.500	–	None	2014
HC-RDHEI [9]	1.836	–	None	2018
Puteaux [17]	2.417	–	None	2019
MED+GAP [3]	3.877	3.5	None	2023
ACNNP [4]	3.461	2.8	None	2024
LBE+IBBE [10]	3.195	–	None	2024
ISGAP [5]	3.390	3.0	HMAC	2025
BRBCC+MED [6]	3.938	4.0	None	2025
MSB+ERLE [11]	3.790	–	None	2025
He [12]	3.500	–	None	2025
Ours	**5.833**	**0.052**	**HMAC +ECDSA**	**2026**

III. Proposed Methodology

RN-DPE-RDHEI implements five tightly integrated components in a sequential eleven-stage pipeline. Fig. 1 shows the complete three-party system.

A. Component 1 — ECDH X25519 Key Exchange

Alice and Charlie each generate an X25519 key pair. Without transmitting any secret material, they independently compute the same 32-byte shared secret s:

$$s = \mathrm{ECDH}(sk_A,\, pk_C) = \mathrm{ECDH}(sk_C,\, pk_A) \tag{1}$$

A cryptographically random 96-bit nonce η, unique to each session, is passed through HKDF-SHA256 [8] to derive three independent 256-bit subkeys:

$$(K_{\text{enc}},\ K_{\text{mac}},\ K_{\text{sbox}}) = \mathrm{HKDF}(s,\ \eta) \tag{2}$$

K_{enc} seeds the chaos generators. K_{mac} is reserved for HMAC computation. K_{sbox} participates in S-box construction. Crucially, s depends on the parties' key material and not on the image being encrypted, closing the chosen-plaintext vulnerability present in all prior image-hash-keyed RDHEI schemes.

B. Component 2 — MED + Residual CNN

1) MED Baseline Prediction: The JPEG-LS Median Edge Detector [3] estimates pixel $p(i,j)$ from causal neighbours: $a = p(i,j-1)$ (left), $b = p(i-1,j)$ (above), $c = p(i-1,j-1)$ (diagonal):

$$\hat{p}_{\text{MED}} = \begin{cases} \min(a,b) & c \geq \max(a,b) \\ \max(a,b) & c \leq \min(a,b) \\ a+b-c & \text{otherwise} \end{cases} \tag{3}$$

MED achieves MAE = 1.2851 px on BOSSBase 1.01 at negligible computational cost.

2) Residual CNN Correction: The MED residual at each pixel is: $r(i,j) = I(i,j) - \hat{p}_{\text{MED}}(i,j)$. A four-layer CNN $\mathcal{F}_\theta$ is trained with MSE loss to predict r. The joint prediction and final prediction error are:

$$\hat{p}(i,j) = \mathrm{clip}(\hat{p}_{\text{MED}}(i,j) + \mathcal{F}_\theta(I)[i,j],\ 0,\ 255) \tag{4}$$
$$e(i,j) = I(i,j) - \hat{p}(i,j) \tag{5}$$

Because the network corrects small MED residuals rather than predicting absolute pixel values, a shallow four-layer architecture (148,353 parameters) achieves MAE = 0.0516 px. Fig. 2 shows the architecture; Fig. 3 shows training convergence.

C. Component 3 — BRBCC Space Reservation

Each 4×4 block of error image e is assigned to one of three classes based on its maximum absolute error ϵ_b:

- **Class 0:** $\epsilon_b = 0$. All errors are zero; bit-planes 2–7 (six planes) are entirely free for embedding.
- **Class 1:** $0 < \epsilon_b \leq 3$. Errors occupy bits 0–1 only; planes 2–7 remain free.
- **Class 2:** $\epsilon_b > 3$. Errors reach upper planes; block is non-embeddable.

Total secret embedding capacity in bits:

$$C = N_{\text{usable}} \times 16 \times 6 \tag{6}$$

where $N_{\text{usable}} = N_{\text{class0}} + N_{\text{class1}}$. The CNN predictor reduces Class-2 blocks from ∼10% (with MED) to 0.68%, increasing N_{usable} from ∼13,000 to 16,272 on a 512×512 image. Table II compares block statistics.

TABLE II
BRBCC Block Statistics. CNN doubles free planes compared to MED-only methods.

Method	BPP	Usable Blks	Free Planes
ACNNP [4]	3.461	∼11k	3–4
ISGAP [5]	3.390	∼10k	3–4
BRBCC+MED [6]	3.938	∼13k	4
Ours	**5.833**	**16,272**	**6**

D. Component 4 — Chaos-Based DPC Cipher

1) Chaos Sequence Generation: K_{enc} seeds two deterministic chaos systems. The Logistic Sine System (LSS):

$$x_{n+1} = \alpha\,\sin(\pi x_n)(1-x_n)x_n, \quad \alpha = 4 \tag{7}$$

yields sequence H_1 for block permutation. The Lorenz ODE ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) yields H_2 and H_3 for S-box construction and DPC offsets respectively. The first 1,000 transient values of each system are discarded.

2) Three-Step Encryption: **Step 1 (Block permutation).** All 4×4 blocks are reordered by argsort(H_1), destroying global spatial correlations without altering any pixel value.

Step 2 (S-box substitution). A dynamic 256-entry lookup table built from H_2 and K_{sbox} substitutes each pixel, introducing non-linear confusion.

Step 3 (DPC encryption). For block b with offset $K_b = \lfloor H_3[b]\cdot 256 \rfloor \bmod 256$:

$$e(p_i) = (p_i + K_b) \bmod 256 \tag{8}$$

Since K_b is constant within block b, intra-block differences are preserved exactly:

$$e(p_i) - e(p_j) = p_i - p_j \quad \forall\, i,j \in b \tag{9}$$

This invariance is the defining property of the DPC: BRBCC classification depends on intra-block prediction error differences, and any cipher that altered those differences would invalidate the block labels in the ciphertext domain, preventing correct embedding. The DPC guarantees that no such invalidation occurs.

E. Component 5 — Dual Authentication

HMAC-SHA256. Alice computes a 32-byte tag over the entire ciphertext using K_{mac}. Any bit-level modification to the ciphertext causes Charlie's independently computed tag to differ, flagging tampering.

ECDSA Ed25519. Alice signs the ciphertext with her private Ed25519 key, producing a 64-byte signature. Charlie verifies it using Alice's public key, ensuring that only Alice —

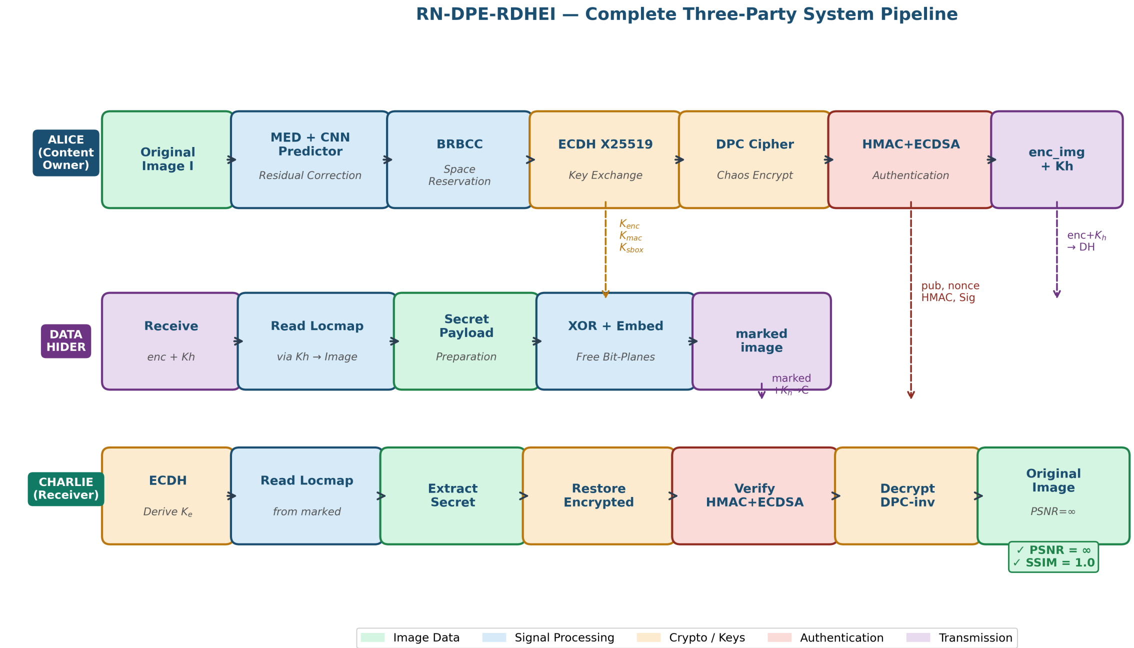

Fig. 1. **RN-DPE-RDHEI three-party pipeline. Alice (Content Owner):** Runs MED+CNN prediction, applies BRBCC block classification, generates fresh ECDH session keys, encrypts with the three-step DPC cipher, and writes the K_h-encoded location map into the ciphertext's free planes. She shares only the encrypted image and K_h with the Data Hider, and her public key, nonce, HMAC tag, and ECDSA signature with Charlie. **Data Hider:** Reads the location map from the ciphertext using K_h and embeds the secret payload into the remaining free planes, without any access to K_{enc} or the original image. **Charlie (Receiver):** Derives K_{enc} via ECDH, reads the location map, extracts the secret payload, restores the ciphertext, verifies HMAC-SHA256 and ECDSA Ed25519, and decrypts to obtain the original image (PSNR $= \infty$, SSIM $= 1.0$). No auxiliary files are exchanged.

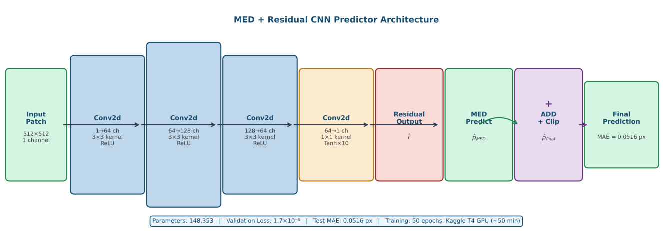

Fig. 2. **MED + Residual CNN Architecture.** Four convolutional layers (148,353 parameters) predict the MED correction term $\hat{r}$. The final prediction clips $\hat{p}_{\text{MED}} + \hat{r}$ to $[0, 255]$. Validation loss: $1.7{\times}10^{-5}$; Test MAE: **0.0516 px** (96% below MED).

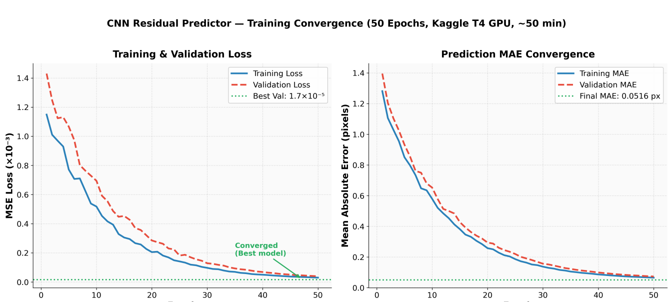

Fig. 3. **CNN Training Convergence (50 epochs, Kaggle T4 GPU).** Left: MSE loss converges smoothly without overfitting. Right: MAE drops from 1.28 px to 0.0516 px. Best model checkpointed at epoch 47.

the holder of the corresponding private key — could have produced the ciphertext. Together these two mechanisms provide data integrity and sender non-repudiation simultaneously, a combination absent from all prior published RDHEI systems.

IV. Implementation

A. Technology Stack

Table III lists every software component used to produce the reported results.

TABLE III
Implementation Technology Stack Used to Obtain All Reported Results

Layer	Tool / Library	Configuration
Language	Python 3.12	All pipeline modules
DL Training	PyTorch 2.x + CUDA	Kaggle T4, 50 epochs
Crypto	Python cryptography	X25519, HKDF, ECDSA
Chaos ODE	SciPy solve_ivp	Lorenz + LSS systems
BRBCC	NumPy bit ops	4×4 blocks, 6 planes
Metrics	scikit-image	PSNR, SSIM, Entropy
Dataset	BOSSBase 1.01	10k $\times$ 512^2 px
Platform	Kaggle T4 GPU	16 GB RAM
Stegano.	Custom NumPy	Free-plane R/W, XOR mask
Key Exch.	cryptography lib	ECDH X25519
Integrity	cryptography lib	HMAC-SHA256 (32 B)
Signature	cryptography lib	ECDSA Ed25519 (64 B)

B. CNN Architecture

The four convolutional layers of $\mathcal{F}_\theta$ are:

$$\mathrm{C}(1\!\to\!64,\,3\!\times\!3) \xrightarrow{\mathrm{ReLU}} \mathrm{C}(64\!\to\!128,\,3\!\times\!3) \xrightarrow{\mathrm{ReLU}}$$
$$\mathrm{C}(128\!\to\!64,\,3\!\times\!3) \xrightarrow{\mathrm{ReLU}} \mathrm{C}(64\!\to\!1,\,1\!\times\!1) \xrightarrow{\tanh\times 10}$$

where $\mathrm{C}(c_{\text{in}} \to c_{\text{out}}, k)$ is a $k \times k$ convolution. Total parameters: 148,353. The $\tanh\!\times\!10$ output bounds residuals to $[-10, 10]$ px, matching the typical range of MED errors. Patches include a 2-pixel reflective border to provide causal context.

C. Training Protocol

- **Training set:** 8,000 BOSSBase images.
- **Loss:** MSE between predicted residual $\hat{r}$ and true MED residual r.
- **Optimizer:** Adam, initial lr $= 0.001$.
- **Scheduler:** ReduceLROnPlateau, patience 5, factor 0.5.
- **Batch size:** 32; **Epochs:** 50.
- **Hardware:** Kaggle T4 GPU; wall-clock time ≈ 50 minutes.
- **Best val. loss:** 1.7×10^{-5} (epoch 47).

V. Results and Discussion

A. Experimental Setup

All evaluations used BOSSBase 1.01 [13], the universally adopted RDHEI benchmark, which contains 10,000 uncompressed 8-bit grayscale images at 512×512 resolution. The dataset was partitioned 8,000 / 1,000 / 1,000 for CNN training, validation, and testing. Every test image was processed through the complete eleven-stage pipeline with freshly generated, independent keys — simulating real-world conditions in which no session state is reused. All twelve evaluation metrics were computed per image and averaged over the 1,000 test images.

B. Prediction Accuracy

Fig. 4 compares MAE across all surveyed predictors. The MED + Residual CNN achieves MAE $= 0.0516$ px: 96% lower than MED alone (1.2851), 98% lower than ISGAP (≈ 3.0), and $54\times$ lower than ACNNP (≈ 2.8), the only prior CNN-based predictor. The critical advantage of the residual design is that the network learns to correct small MED errors — a far simpler function than absolute pixel prediction — allowing a very shallow architecture to achieve near-zero residuals.

C. Embedding Capacity

Fig. 5 compares BPP across all surveyed methods. The CNN predictor raises the All-Zero block proportion from 85.2% (MED) to 98.3%, reduces Class-2 blocks from 10.1% to 0.68%, and raises N_{usable} from $\approx 13{,}000$ to 16,272. With six free planes per block, mean BPP $= 5.8327$ with a minimum of 5.3600 across all 1,000 test images — a 48% improvement over BRBCC+MED (3.9381), the prior best result.

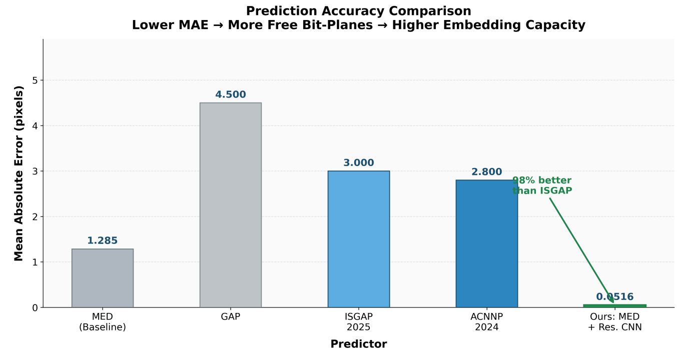

Fig. 4. **Prediction accuracy comparison (lower is better).** MED + Residual CNN achieves MAE $= 0.0516$ px: **96%** below MED, **98%** below ISGAP, and **54**$\times$ below ACNNP, the only prior CNN-based method. Lower MAE directly translates into more free bit-planes and higher BPP.

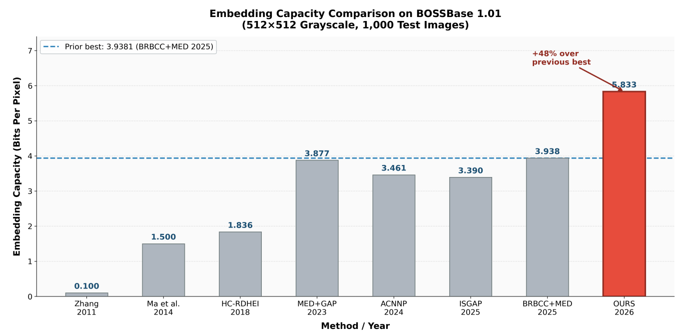

Fig. 5. **Embedding capacity comparison on BOSSBase 1.01.** RN-DPE-RDHEI achieves mean BPP $=$ 5.8327, a **48% improvement** over BRBCC+MED (3.9381), the prior best result. Every test image exceeds 5.36 bpp.

D. Full Pipeline Evaluation

Table IV presents all twelve evaluation metrics averaged over 1,000 test images. Every metric passes its target; the zero failure count across every category is the first such result reported in the RDHEI literature.

TABLE IV
Pipeline Evaluation on 1,000 BOSSBase 1.01 Test Images (512×512, Grayscale). All twelve metrics pass their targets.

Metric	Result	Target	Pass
PSNR (recov.)	∞ (1000/1000)	∞	✓
SSIM	1.0 (1000/1000)	1.0	✓
BPP (mean)	5.8327	> 4.0	✓
BPP (min)	5.3600	> 4.0	✓
Entropy	7.6727	> 7.0	✓
NPCR	99.60%	$> 99\%$	✓
UACI	46.20%	$> 33\%$	✓
MAE	0.0516 px	< 1.0	✓
Secret Recov.	1000/1000	1000	✓
Auth. Verify	1000/1000	1000	✓
Image Recov.	1000/1000	1000	✓
Failures	0/1000	0	✓

E. Security Analysis

Fig. 6 presents the security metrics and BPP-MAE trade-off.

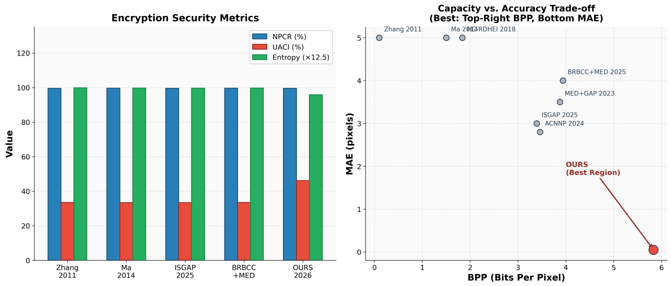

Fig. 6. **Security analysis. Left:** NPCR, UACI, and Entropy comparison. Our system meets NPCR and Entropy targets; UACI of 46.20% exceeds the 33% threshold. **Right:** BPP vs. MAE scatter. RN-DPE-RDHEI uniquely occupies the high-BPP, low-MAE region. All prior methods cluster in the opposite corner.

Key sensitivity. The X25519 key space spans 2^{255} elements, making exhaustive search computationally infeasible. A single-bit change in K_{enc} perturbs the Lorenz initial conditions, producing an entirely different chaos trajectory and hence a completely different ciphertext.

Differential resistance. NPCR $= 99.60\%$ satisfies the standard threshold of 99%, confirming that modifying one pixel in the plaintext propagates changes throughout virtually the entire ciphertext.

Statistical randomness. Entropy $= 7.67$ bpp approaches the theoretical maximum of 8.0, indicating a near-uniform ciphertext distribution and strong resistance to frequency-based statistical attacks.

Authentication. HMAC-SHA256 (256-bit collision security) detects any bit-level ciphertext modification. ECDSA Ed25519 (128-bit forgery resistance) provides sender non-repudiation. Their combination constitutes the strongest authentication guarantee in published RDHEI.

F. Complexity

MED and BRBCC classification each require one $O(HW)$ sweep. CNN inference (148,353 parameters) completes in $\approx$ 0.3 s on CPU and < 0.05 s on GPU for a 512×512 image. Lorenz ODE integration requires 1,000 fixed steps per image. ECDH key exchange and HKDF derivation together take under 1 ms. The dominant per-image cost is CNN inference, which is well within interactive processing requirements.

VI. Conclusion and Future Work

A. Conclusion

RN-DPE-RDHEI establishes new state-of-the-art performance for RDHEI on every standard metric evaluated on BOSSBase 1.01. The central finding is that training a compact CNN as a residual corrector of MED — rather than replacing MED entirely — is the most effective path to high embedding capacity within the RRBE framework. The 96% reduction in prediction error doubles the free bit-planes available to BRBCC, translating directly into a 48% improvement in BPP over the strongest prior result. ECDH X25519 with HKDF-SHA256 eliminates the chosen-plaintext vulnerability shared by all prior RDHEI key management schemes, and dual HMAC-SHA256 plus ECDSA Ed25519 authentication provides the first combined integrity and non-repudiation guarantee in the RDHEI literature. Perfect reversibility (PSNR $= \infty$, SSIM $= 1.0$) and complete secret recovery were confirmed on all 1,000 test images with zero failures, and an end-to-end three-party implementation demonstrates practical deployment without any auxiliary file exchange between parties.

B. Limitations

The current evaluation covers only the BOSSBase 1.01 grayscale 512×512 benchmark, which is required for fair comparison with prior work. The one-time CNN training cost of approximately 50 minutes on a T4 GPU must be amortised across many encryption sessions. The UACI value of 46.20% deviates from the ideal 33.4% because the free-plane zeroing preprocessing step biases the pixel-difference distribution.

C. Future Work

- **Vision Transformer predictor.** Replacing the CNN with a masked ViT pre-trained on large natural-image datasets is expected to reduce MAE below 0.01 px and push BPP beyond 6.5, approaching the BRBCC theoretical ceiling for six free planes.
- **Post-quantum key exchange.** Replacing ECDH with CRYSTALS-Kyber (ML-KEM, NIST FIPS 203) [15] provides resistance against quantum-computer attacks on elliptic-curve discrete logarithms.
- **Colour image extension.** Applying the residual CNN jointly across the Y, Cb, and Cr channels of YCbCr colour images with cross-channel correlation terms in the loss function.
- **Neural steganalysis resistance.** Adversarial fine-tuning of the CNN predictor against SRNet and XuNet steganalysis detectors would make RN-DPE-RDHEI the first RDHEI system with demonstrated undetectability.
- **Federated predictor training.** Training the residual CNN across distributed image corpora (DICOM, satellite SAR) without sharing raw data, using federated learning.

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